MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) MODEL QUESTION PAPER

FOURTH SEMESTER EXAMINATION

MG4DSCPEG200- ATF AND LATEST DEVELOPMENT IN FUEL TECHNOLOGY

Duration: 1.5 hrs. Maximum marks: 50

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer **All** Questions

Each question carries **1** mark

1. Identify the source material from which biofuels are derived. [Remembering][CO1]

   A) Fossil reserves B) Biomass
   C) Metallic ores D) Synthetic polymers

2. Recognize the term used for ethanol blended with gasoline at 85%. [Remembering] [CO2]
   A) E10 B) E20
   C) E85 D) E5

3. Associate ethanol's high-octane number with its engine effect. [Understanding] [CO2]

   A) Increased knocking B) Reduced knocking

   C) Increased sulfur emission D) Increased particulate matter

4. Aviation Turbine Fuel (ATF) obtained from crude oil mainly belongs to which fraction? [Understand] [CO3]
   A) Gasoline fraction B) Kerosene fraction

   C) Diesel fraction D) Residual oil

5. One major advantage of SAF over conventional ATF is: [Remember] [CO3]

   A) Higher sulphur content B) Higher freezing point
   C) Reduction in greenhouse gas emissions D) Lower energy density

6. The copper corrosion test in aviation fuels is used to determine the presence of: [Remember] [CO4]
   A) Aromatic content B) Corrosive sulphur
   C) Water content D) Smoke tendency

7. High aromatic content in ATF leads to which of the following. [Understand] [CO5]

A) Better combustion B) Increased smoke

C) Lower density D) Reduced freezing point

8. What is the main reason for using conductivity additives in aviation fuels? [Remember][CO5]

A) Improving combustion B) Preventing static charge accumulation

C) Reducing smoke D) Increasing calorific value

9. Why are second-generation biofuels considered more sustainable than first-generation biofuels? [Understand] [CO6]

A) Require high fertilizer input B) Use non-food biomass and wastes

C) Have high freezing points D) Increase carbon emissions

10. Why is fuel lubricity considered a success criterion for aviation biofuels? [Understand][CO6]

A) It improves combustion efficiency B) It prevents corrosion of fuel tanks

C) It protects fuel pumps and engine components D) It increases octane number

(10x1=10)

Part B

Short Essay Type Questions

Answer Any **4** Questions

Each question carries **3** marks

11. Explain the renewable nature of biofuels. [Understanding][CO1]
12. List any four important fuel tests used for ethanol fuels. [Remembering] [CO2]
13. List any three important physico-chemical properties of aviation fuels. [Remember][CO3]
14. Explain the significance of the copper corrosion test for aviation fuels. [Understand] [CO4]
15. Discuss the Merox process? State its purpose in jet fuel refining. [Understand] [CO5]
16. Differentiate between first- and second-generation biofuels. [Understand][CO6]

(4x3=12)

Part C

Essay Type Questions

Answer Any **4** Questions

Each question carries **7** marks

17. Explain the environmental sustainability of biofuels by summarizing benefits, concerns, and the role of Life Cycle Assessment (LCA). [Understanding][CO1]
18. Describe dry milling technology for ethanol production with key steps and co-product. [Remembering] [CO2]
19. Describe the technology of aviation fuel preparation before application to aircraft tanks. [Understand] [CO3]
20. Illustrate anti-knock performance of aviation gasoline and describe Research Octane Number and Motor Octane Number. [Apply][CO4]
21. Outline the hydrodesulphurization process used in aviation fuel production. [Remember] [CO5]
22. Explain second-generation biofuels for aviation. Discuss their feedstocks, production technologies, and applications. [Understand] [CO6]

(4x7=28)